Examining the Efficacy of Mitigation Strategies Against White-Nose Syndrome (WNS) Using Mathematical Methods

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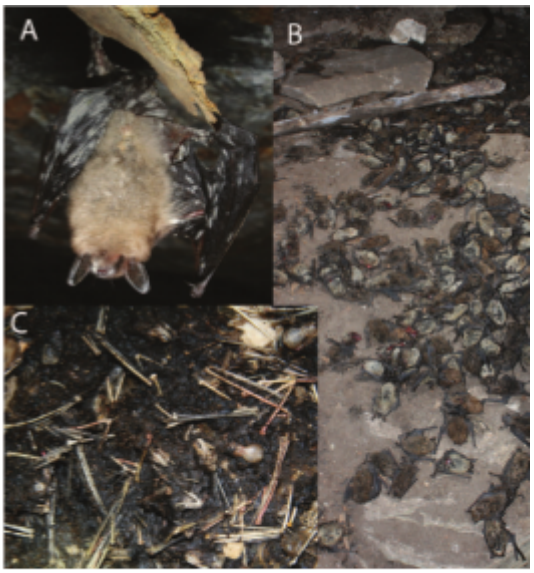
Introduction

White-nose syndrome (WNS) is a disease that has silently eliminated significant portions of hibernating bat populations across North America since its discovery in 2006. It's caused by the psychrophilic, meaning that it's cold-loving, fungus Pseudogymnoascus destructans, and it has killed over 6.7 million bats; some species even experienced mortality rates exceeding 90% (Frick et al. 2010). The removal of bats from the ecological cycle has immense consequences not just in terms of the environment but also in terms of the economy: they play a large role in pest suppression, pollination, and seed dispersal, and thus their increased mortality can lead to billions of dollars being lost in the agricultural industry (Kunz et al., 2011). There are reports of the disease in new hibernacula or caves each year, and WNS continues to spread through North America, affecting 40 U.S. states and seven Canadian provinces, despite extensive research conducted on the disease (Hoyt et al., 2021). Environmental reservoirs, like soil and sediment in the cave, are the main sustainers and drivers of the fungus, because WNS is not able to spread without it present (Hicks et al., 2023). There is an immediate urgency for effective mitigation strategies, as a result of high rates of mortality, widespread persistence of the P. destructans fungus, and the importance of bats within the environment. Therefore, I propose an important question to explore: how can a new compartmental model be used to explore the efficacy of mitigation strategies against White-Nose Syndrome (WNS) within cave-hibernating bat populations?

There are challenges that are associated with WNS, and said challenges stem from the pathogen's biology and the vulnerability of the bats. Because P. destructans's ideal environment is cold and humid, that of a typical hibernacula, the bats residing in the hibernacula during torpor with lowered body temperatures are especially susceptible to the infection (Verant et al., 2018).

This infection, however, also disrupts bat hibernation cycles, which eventually causes depletion of fat reserves and ultimately death as a result of starvation (Langwig et al., 2017). There are limitations to all of the current mitigation strategies, like fungicide treatments, microclimate manipulation, and ultraviolet light exposure (Duan et al., 2021). Despite each strategy targeting specifically a singular aspect, there is not a single mitigation strategy that has proved to be the most effective; however, these strategies coupled together could, in theory, effectively eliminate WNS from any bat population. There has been a lack of research on the comparative efficacy of the strategies in a single hibernacula, thus making it difficult for effective mitigation of WNS.



(Langwig et al., 2017)

Previously, research has shown the different parts of WNS in caves that complicate the problem, like the cold environmental conditions of P. destructans, the disease's transmission from bat to bat and from the environment to bats, and the physiological aspect of the bats that are infected (Haase et al. 2020; Verant et al., 2018). The study conducted by Duan et al. (2021) modeled the potential interventions to assess their efficacy with regard to bat dispersal between two hibernacula. In addition, Hicks et al. (2023) demonstrated the need for interventions, specifically those that target fungal persistence by showcasing how these environmental reservoirs are important in facilitating the presence of the P. destructans fungus. On the other hand, Langwig et al. (2017) suggest that vaccination and other host-targeted strategies should also have significant consideration, as they concluded that only certain bat populations possess resistance mechanisms that are adaptive. The issue with these studies is that they often isolate

one aspect of the disease, which leads to significant gaps in the understanding of the efficacy of each mitigation strategy. Whereas, simulated environments allow for unconstrained experimentation that has the potential to provide unforeseen solutions to the disease.

Even though the current research is important, there is nevertheless a lack of research that explicitly compares the efficacy of several mitigation strategies under a simulated environment that is controlled and standardized. As discussed earlier and as will be discussed in the review of literature, there has been research assessing specific interventions, but said mitigation efforts have not been analyzed in comparison to one another using parameters that are kept consistent to assess their relative effectiveness. By addressing this gap, I hope to aid in the discussion of mathematical modeling and environmental conservation, specifically assisting in the prioritization of interventions for WNS.

The purpose of my study is to create and propose a compartmental model that assesses the efficacy of three important mitigation measures: vaccination, fungicide treatment, and microclimate manipulation. I hope to adjust the model parameters representing these mitigation measures individually while keeping other variables constant to assess how each intervention impacts the dynamics of the disease. By doing so, I'll be able to shed light on the conditions under which different strategies are effective, contributing to our understanding of what is required to reintroduce bats to their natural hibernacula.

My research has significant implications for the scientific community. Firstly, I address a gap in the literature by directly using the proposed compartmental model to compare mitigation strategies, thereby providing practical insights into how to optimize resources for the intervention. These conservation methods are vital in terminating WNS, given the fact that bats are essential for the environment and the economy. Next, by adding new compartments into the

standard SIR (Susceptible-Infected-Removed) model, my research will showcase the ability of mathematical models to evaluate complex systems, and whether it's worthwhile to continue using these models in general.

Literature Review

White-Nose Syndrome (WNS) and the environmental reservoirs that facilitate it

The main reason why White-Nose Syndrome (WNS), which is caused by the fungal pathogen Pseudogymnoascus destructans, is very prevalent in bat hibernacula is because of its ability to persist in environmental reservoirs, like soil and sediment. In this section, I will analyze existing literature to showcase the various factors in the environment that facilitate fungal persistence and their consequences for WNS management strategies.

To investigate the sustainability of P. destructans in the environment without the presence of infected bats, Hicks et al. (2023) translocated 79 uninfected Little Brown Bats (Myotis lucifugus) to two contaminated hibernacula; then, in the period of eight weeks, some of the bats were severely infected and some were beginning to die by week 14, illustrating that the environmental reservoirs alone, in this case the P. destructans fungus, could initiate and sustain the outbreaks of WNS. As a result, it's evident that mitigation strategies are instrumental, and that this aspect should be incorporated when seeking to minimize disease propagation. However, limitations of the study include the stress of translocation that could have caused unusual behavioral manifestations, in addition to the limited geographic scope.

In a similar manner, Verant et al. (2018) researched the temporal and spatial distribution of the fungus P. destructans in the hibernacula. They found that sediment was an important environmental reservoir for the persistence of WNS and that there were higher fungal loads in the sediment than wall surfaces by sampling six hibernacula with various progressions of WNS.

Since the detection of WNS, they also concluded that temperature and time were significant in predicting the prevalence of the fungus, highlighting the resiliency of the P. destructans fungus in specific microclimates, but the study is limited by its small sample size and lack of empirical experimental validation for the viability of the fungus.

Continuing the research within the same scope, Torres-Cruz et al. (2019) extended the work of Verant et al. (2018) by analyzing microclimate conditions in ten caves in New Mexico to determine whether they can suit the growth of P. destructans. In their findings, they noted that eight of the 10 caves had optimal conditions for the proliferation of the fungus: specifically, in temperatures between 3-16 °C and humidity levels that exceed 90%. Despite the favorable conditions in all of the caves examined, P. destructans was not present in the area, leading to the conclusion that environmental suitability does not guarantee fungal presence. However, there could have been a discrepancy in the study's static measurements of temperature and humidity levels, causing an underestimation of short-term microclimatic variability.

As a whole, these studies highlight the importance and the conditions of environmental reservoirs in sustaining WNS. The results of Hicks et al. (2023) and Verant et al. (2018) emphasize the resilience of P. destructans in sediment, while on the other hand Torres-Cruz et al. (2019) emphasize the importance of the microclimate as a key determinant of the resilience of the fungus. A common theme in each of these studies is the need for environmental interventions and mitigation strategies to disrupt the pathogen's life cycle. Certainly, there are varying degrees of reliability in assessing fungal persistence, suggested by the differences in methodologies, like direct experimentation versus observational. By understanding the nuances of each study, it becomes clear that targeting environmental reservoirs, like those that aid in the persistence of WNS, is essential for effective management of WNS. Not only do the studies demonstrate the

limitations of current research in accounting for the variability in the environment, but they also provide valuable insights into the specific conditions that facilitate fungal survival.

How factors that are host-specific affect the dynamics of WNS

Host physiology, behavior, immune responses, and other host-specific (bat-specific) factors greatly impact the progression of and the outcomes of WNS. By analyzing the host-specific factors, I hope to better understand the few principal factors that contribute to the variability in disease susceptibility and resistance, specifically that of WNS.

In their study, Haase et al. (2020) investigated how body mass, torpor metabolic rate, and evaporative water loss affect the survival probability of bats. Through the integration of field data with individual-based hibernation energetics models, they found that smaller bats and bats in humid environments exhibited higher mortality rates, because the optimal environment led to increased fungal growth, thus leading to excessive energy expenditure. Although the study's simplified model, which ignores immune responses, limits the comprehensiveness, their findings suggest that physiological traits are important for determining WNS outcomes.

Similarly, Langwig et al. (2017) explored the resistance mechanisms in persisting Little Brown Bat populations. By leveraging field observations and mathematical models, they found that bats in surviving populations possessed significantly lower fungal loads, concentrations of the P. destructans fungus, compared to bat populations that were in decline. The researchers attributed this resistance to adaptive changes, like generally reduced fungal growth and potential adaptations of the bats' immune systems. It's difficult to isolate genetic versus environmental drivers of this resistance, because of the study's reliance on cross-sectional data across different populations.

A review conducted by Isidoro-Ayza et al. (2024) had a focus on the pathogenesis of WNS and the role of immune response in the mitigation of disease progression; specifically, their review synthesized findings from in vitro studies' experimental infections and genomic analyses of the P. destructans fungus. The study confirmed the nuanced immune responses of bats by concluding that the host immune responses are highly variable across different species, where some bats endured WNS and others failed to limit the fungal growth. This study provides a comprehensive understanding of host-pathogen interactions with regards to WNS because of its broad scope; consequently, it lacks specific information regarding intervention strategies.

Taken together, these studies reveal that host factors, such as body mass, metabolic rate, and immune responses play an important role in the dynamics of WNS. While Isidoro-Ayza et al. (2024) underscore the variability of immune responses of each bat, Haase et al. (2020) and Langwig et al. (2017) emphasize physiological and adaptive traits. It's commonly seen in the literature that interventions should consider host-specific characteristics, like vaccinating immunologically weaker populations. These studies can inform mathematical models to incorporate host variability or a host compartment, significantly revolutionizing the current environmental reservoir models. Despite the research providing a foundation for understanding host-pathogen interactions, there are gaps that remain in linking the physiological and immunological data with intervention outcomes. By addressing these gaps, we can enhance the precision of the models used to inform mitigation strategies.

Mathematical models in understanding dynamics of disease

Mathematical models are essential for understanding WNS transmission dynamics and assessing the efficacy of mitigation strategies. In this theme, I will utilize existing literature to demonstrate how models have been applied to evaluate interventions and identify the limitations of the current mathematical approaches.

Duan et al. (2021) developed a compartmental model to analyze bat dispersal by comparing five mitigation strategies: vaccination, fungicide treatment, microclimate manipulation, soil bacteria application, and UV light exposure. They found that vaccination was the most effective by simulating the effects of these interventions over 10 years, where they found that vaccination, at low implementation intensities, led to the survival exceeding 50%. This study raises concerns about the real-world applicability of its results because they assumed idealized conditions, which is difficult to come across realistically.

Applying compartmental models to coronavirus, Menda et al. proposed reactive SEIRD (R-SEIRD) models to address limitations in standard compartmental models, like fixed infection rates. The authors clearly showed the improved accuracy in predicting outbreaks with several peaks by incorporating dynamical parameters. While the disease chosen was COVID-19, the methods of the study are relevant for WNS modeling, specifically in capturing the dynamic seasonal variations in fungal infections. There are challenges in interpretability and computational complexity of the study as a result of relying on neural networks.

Focusing on specifically bat populations, Weins et al. (2023) presented a demographic modeling tool for projecting their dynamics under multiple stressors, which they named BatTool. The tool showcased its utility in assessing the trajectories of populations by applying it to a case study involving wind energy development. While not specific to the fungus P. destructans, the BatTool's framework can be used to compare to other models, since it's supported by the U.S.

Fish and Wildlife Service, and can be adapted to evaluate WNS interventions alongside other methods. It does, though, obscure localized variations because of its reliance on aggregated data.

Collectively, these studies highlight the potential applications and limitations of mathematical models in assessing WNS mitigation strategies. Duan et al. (2021) and Wiens et al. (2023) focus on the models that are intervention-specific; Menda et al. (2021) advocated for methodological advancements. Although their assumptions and even simplification often limit the applicability in a complex environment, mathematical models allow for the comparison of interventions by providing a quantitative basis. To improve their guiding of conservation strategies, future models should incorporate environmental variability, specific traits, and other logistical constraints.

Conclusion of literature review

This review of literature included the identification of significant environmental and physiological determinants of WNS outcomes and demonstration of the practicality of mathematical models in the evaluation of interventions. The strengths of the literature analyzed are in the interdisciplinary nature and the integration of different empirical and theoretical approaches. However, as with any research, there are limitations, including reliance on conditions that are ideal, limited sample sizes, and simplified assumptions in the models.

Among the gaps in the literature is the lack of research applying advanced mathematical models, like SIRP (Susceptible, Infected, Recovered, and Pathogen concentration), to assess the mitigation strategies within a single cave. Moreover, most models do not integrate environmental data from real-time updates and variability in bat-specific immune responses. My research seeks to address the first gap stated, as I do not have access to real-time data or a cave. I propose a

compartmental model that assesses the efficacy of vaccination, fungicide treatment, and microclimate manipulation in single-cave systems. I hope to contribute to a more nuanced understanding of WNS mitigation strategies and provide practical insights for mitigation efforts.

Methodology

By proposing and analyzing a mathematical compartmental model, the purpose of this study is to explore the extent to which various mitigation strategies affect the epidemiological growth of White-Nose Syndrome (WNS) within cave-hibernating bats. Under the guidance of Professor Erin Landguth, whose research primarily focuses on computational epidemiology, I propose a compartmental mathematical model through ordinary differential equations to explore this issue. Unlike qualitative methods that focus on human accounts and opinions, this study requires mathematical precision. Also, mathematical modeling facilitates the comparison of multiple mitigation strategies, specifically under controlled conditions. Likewise, quantitative methods are crucial in epidemiological research as they allow for precise estimation of the dynamics of disease and the outcomes of the implemented mitigation strategies (Creswell, 2014).

Because a mathematical analysis provides a means of simulating the dynamics without ethical and logistical complications of direct field experimentation, I was able to run hundreds of simulations of intervention strategies, under controlled conditions, that would be virtually impossible to test in current bat populations due to concerns regarding conservation, which consequently leads to a limitation in data collection. By setting up a quantitative framework, I'm thus able to provide calculated, key thresholds in disease transmission, such as the basic reproduction number (R_0), which characterizes the ability of the disease to spread. Additionally, by using compartmental modeling, I ensure that the research is not only structured but also directly informs our understanding of WNS transmission and dynamics, aligning with the best

practices in the field (Leedy and Ormrod, 2019). Compartmental modeling provides the advantage of being able to conduct sensitivity analyses that ascertain which parameters are most detrimental to bat populations. Integrating that with the flexibility of these models to capture disease dynamics, the result is a framework that is adaptable to new empirical conclusions, which makes them especially effective for long-term conservation initiatives.

The main tool that I used was a compartmental SIRP (Susceptible, Infected, Recovered, and Pathogen concentration) model, which is widely implemented in diseases that stem from fungi that are facilitated by various environmental reservoirs such as soil, water, etc. In fact, the SIRP model was specifically designed to address the inclusion of the environment in the field of mathematical epidemiology (Giménez-Romero et al., 2021). Since I didn't have access to empirical data that met the criteria for my research, I simply used theoretical estimates that were based on acknowledged epidemiological principles. The table below summarizes my primary model parameters.

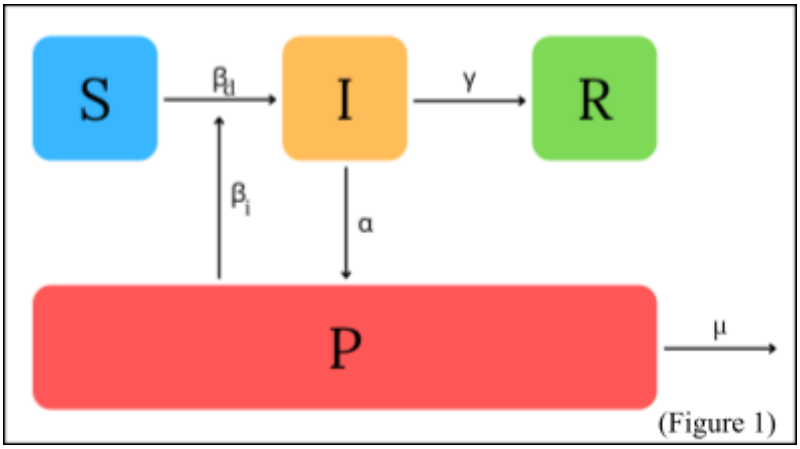
Definition	**Parameter**
Direct transmission rate (from bat to bat)	β_d
Indirect transmission rate (from the pathogen)	β_i
Host mortality rate	γ
Host shedding rate of pathogen	α
Pathogen decay rate	μ

The study focused on cave-hibernating bat populations that were affected by WNS, without narrowly focusing on any specific species. In what follows, I describe how we defined the SIRP model, constructed the next-generation matrix, derived the basic reproduction number R_0, and conducted analyses to holistically examine the impact of each parameter on R_0. It's important to note that the following

steps were significantly supported by Professor Erin Landguth with feedback from Professor Jon Graham of the University of Montana.

Procedure

1. Define the SIRP model compartments and derive the equations that govern their linkage; the resulting differential equations were established based on epidemiological modeling practices (Giménez-Romero et al., 2021). These equations account for the dynamics of the bat populations and the pathogen persistence in the environment, which further assists in structurally analyzing the mitigation strategies.



(Figure 1)

Figure 1 provides a visual representation of the SIRP compartmental model. In this model, the S represents the compartment containing the susceptible bat population, I represents the infected bat population, R represents those that have died (removed from the population), and P is the pathogen within the environment. The derived differential equations (Figure 2) describe the dynamics of a cave with a bat species, reflecting biologically realistic disease transmission routes and fungal persistence that enabled me to assess various intervention strategies. They also capture how each compartment contributes to the overall system and, as discussed later, how each parameter contributes to the basic reproduction number. As an example, the first equation describes the

$$\frac{dS}{dt} = -\beta_d SI - \beta_i SP$$
$$\frac{dI}{dt} = \beta_d SI + \beta_i SP - \gamma I$$
$$\frac{dR}{dt} = \gamma I$$
$$\frac{dP}{dt} = \alpha I - \mu P$$
(Figure 2)

change in the susceptible bat population as a function of interactions between susceptibles and infected bats (SI) and susceptibles with the environment (SP). Note that both terms are negative, indicating that these interactions effect a decrease in the susceptible population. The parameters β_d and β_i represent the rates of susceptible population decline due to these interactions. Other equations can be similarly interpreted.

2. Construct the next-generation matrix (NGM) to determine the key factors that drive the disease. To isolate the system that drives the population's infection, we need to reduce the SIRP system mathematically, an important step to reduce the complexity of the model while maintaining the main drivers.

In order to analyze the dynamics of the disease in bat populations, we used the Next Generation Matrix (NGM) to calculate the average number of new infections caused by a single infected bat. Following the Diekmann et al. (1990, 2010) method, we divide the SIRP model into two parts; one for new infections, including indirect or direct transmission, and one for transitions, like death. This is given by:

$$J = T + \Sigma$$

where T and Σ are matrices representing the infection and transmission sections, respectively. And, to determine the NGM, represented by K, we used the following equation:

$$K = T\Sigma^{-1}$$

After completing the derivation of K, we got:

$$K = \begin{bmatrix} \frac{\beta_d}{\gamma} + \frac{\beta_i \alpha}{\gamma \mu} & \frac{\beta_i}{\mu} \\ 0 & 0 \end{bmatrix}$$

3. Derive R_0, the basic reproduction number, as a function of key model parameters. The basic reproduction number R_0 is essentially the threshold to determine whether WNS can spread and the speed at which it will spread. It turns out that R_0 is equivalent to the largest eigenvalue of the NGM (Diekmann et al. 1990; 2010).

After taking the determinant of the NGM, the largest eigenvalue $\lambda = R_0$ is computed as:

$$\lambda = R_0 = \frac{\beta_d}{\gamma} + \frac{\beta_i}{\gamma} * \frac{\alpha}{\mu}$$

Finding R_0 in terms of each parameter is particularly important, since, now, one is able to analyze the roles of each parameter in shifting R_0. In other words, this equation coupled with statistical analyses can be used to determine the efficacy of mitigation strategies in reducing R_0, or to help reduce and eventually remove WNS disease from the cave.

4. Conduct several sensitivity analyses: use a one-at-a-time approach to study the impacts of each parameter individually, construct 3D surface plots to demonstrate these impacts visually, and conduct a Normalized Relative Sensitivity (NRS) analysis.

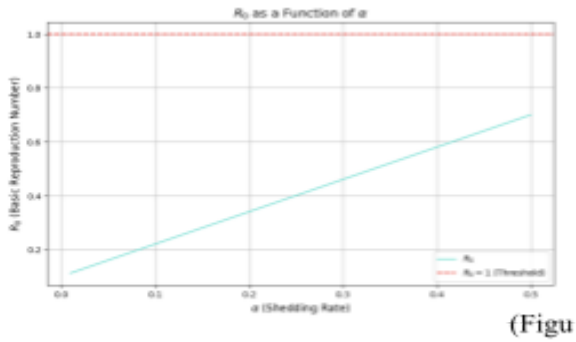
For the one-at-a-time approach, I fix four of the five model parameters and vary one, and then plot R_0 as a function of the parameter varied. Using 3D surface plots, I examine the impact of two parameters simultaneously on R_0. Finally, the NRS analysis provides a numerical

approximation of each parameter's influence on R_0. These analyses are elaborated upon in the succeeding section. See Appendix A for all of the corresponding Python Code.

Results

One-at-a-time Analysis with SIRP parameters.

I conducted a one-at-a-time sensitivity analysis on each parameter in my SIRP modeling, meaning that I varied one parameter at a time while holding the others constant to observe its impact on the R_0 value, the basic reproduction number. I analyzed all of the parameters, which are β_d (direct transmission rate from bat to bat), β_i (indirect transmission rate from environment to bat), γ (host mortality rate), α (shedding rate by the infected hosts), and μ (pathogen decay rate). Using a Python library for numerical and scientific computing called NumPy, I was able to use their `linespace()` function to set an interval for each parameter from 0.01 to 0.5 with a resolution of 100, meaning 100 values were evenly spaced and calculated over that interval. Also, I used another Python library called Matplotlib, which is often used in data analysis, scientific research, and engineering for data visualization, to create a plot with the corresponding R_0 values, using a dashed line to indicate the threshold value of $R_0 = 1$.



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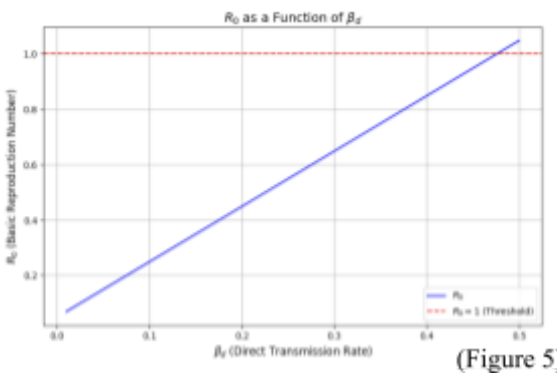

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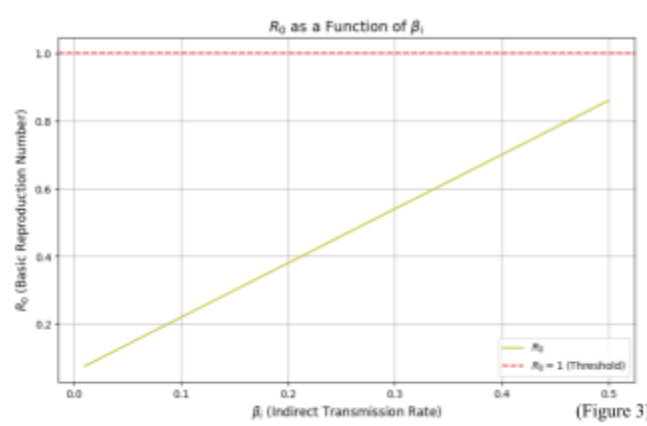
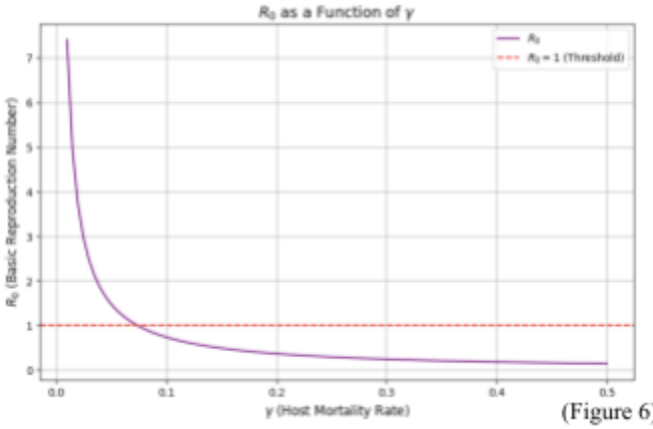


(Figure 3)

After hundreds of simulations and fixing code bugs, I came to find two distinct trends. There was a linear increase in R_0 as a function of the parameters β_d (Figure 5), β_i (Figure 3), and α (Figure 4), showing that higher transmission rates, including direct and indirect, and shedding rates directly correlate with disease persistence, as would be expected. Alternatively, γ (Figure 6) and μ (Figure 7) illustrated an exponential decay relationship with R_0, showing that increasing these parameters will exponentially reduce disease persistence. The results showcase that strategies that aim at decreasing host mortality are actually increasing R_0, which is logical since a deadly disease is removed from a system faster, as the number of infected hosts reduces over time. Reducing transmission and shedding rates seems to effectively control the bat populations over a long period of time.



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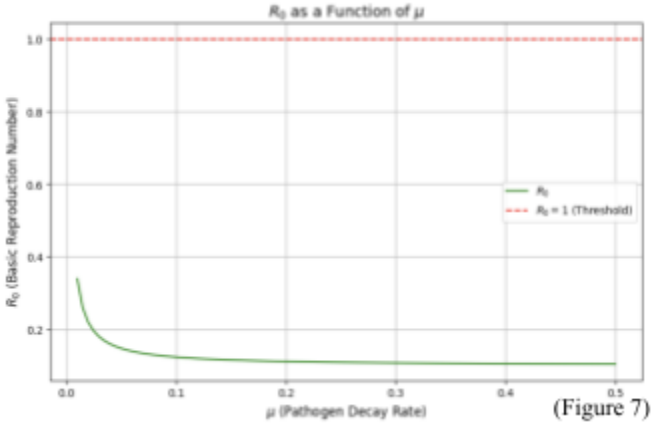
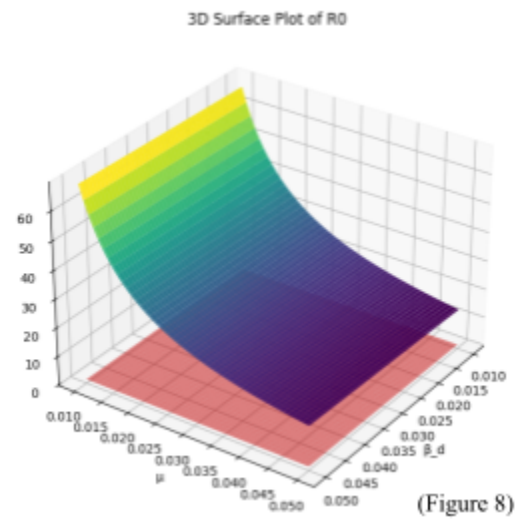


(Figure 7)

Through the numerical simulation above, increasing β_d from 0.01 to 0.5 resulted in the R_0 increasing linearly from 0.068 to 1.048, nearly a fifteen fold increase, illustrating the direct proportionality between β_d and R_0 (Figure 5). Likewise, increasing β_i or α led to similar linear growth (Figures 3 and 4), though the magnitude at which R_0 increases is noticeably different. However, R_0 decreased exponentially with increasing γ and μ (Figures 6 and 7).

3D Surface Plots Through Multi-parameter Sensitivity Analysis

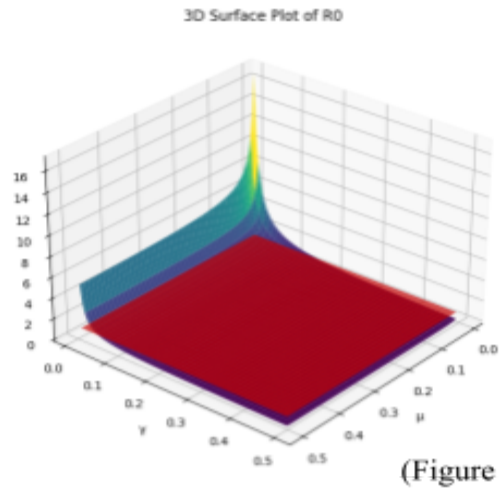
To further investigate the relationship between parameters and their overall influence on the R_0 function, I created contour plots and 3D surface plots to examine the joint effects of parameters on R_0. I did this by using the same NumPy and Matplotlib libraries, but the code definitely had some alterations. In addition to the aforementioned `linespace()` function, I used the NumPy `meshgrid()` function, which simply takes in the ranges of two parameters and copies them for each row (if it'll be the x-axis) or for each column (if it'll be the y-axis) to create 2D matrices to allow for the contour and 3D plotting. I then recalculated R_0 with these new parameter meshgrids to create a R_0 meshgrid to plot on the Z-axis while keeping the other parameters constant. Also, I came across a program error that prompted me to ensure that level values are only increasing, and I made sure to include a red plane to indicate the $R_0 = 1$ threshold. I analyzed three parameter relationships: μ vs. β_d vs. R_0, β_i vs. β_d vs. R_0, and γ vs. μ vs. R_0. The motivation for choosing these parameters was to compare the parameter that yielded an R_0 of exponential decay to linear, linear to linear, and exponential decay to exponential decay.



(Figure 8)

For the first comparison of μ (pathogen decay rate) vs. β_d (direct transmission rate) vs. R_0, the β_d values, which ranged from 0.01 to 0.5, didn't have much effect on R_0 compared to increasing μ (Figure 8). While lowering β_d helps, increasing μ should definitely be a priority.

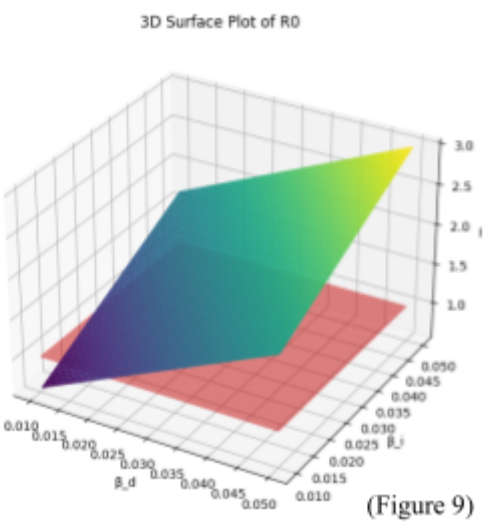
This comparison clearly indicates that increasing the decay rate of the *P. destructans* pathogen is really effective, but decreasing the direct transmission still has an observable effect.



(Figure 10)

As for the second analysis, which compared β_i (indirect transmission rate) vs. β_d (direct transmission rate) vs. R_0, the results showed that to achieve an effective mitigation strategy (keeping $R_0 < 1$) will require keeping β_i below 0.25 and β_d below approximately 0.21 (Figure 9). This showcases the importance of controlling both direct and indirect transmission rates to mitigate WNS effectively.

In Figure 10 displaying γ (host mortality rate) vs. μ (pathogen decay rate) vs. R_0, I found that increasing γ had a stronger effect on reducing R_0 than increasing μ. However, there is more nuance to it. Higher γ leads to more bats dying, which is what we're trying to avoid, and it can lead to further population decline. The results showcase that while μ has an impact, it's less significant than γ. For effective control of WNS, γ should be kept around 0.15, thereby not leading to excessive deaths while controlling the disease properly.

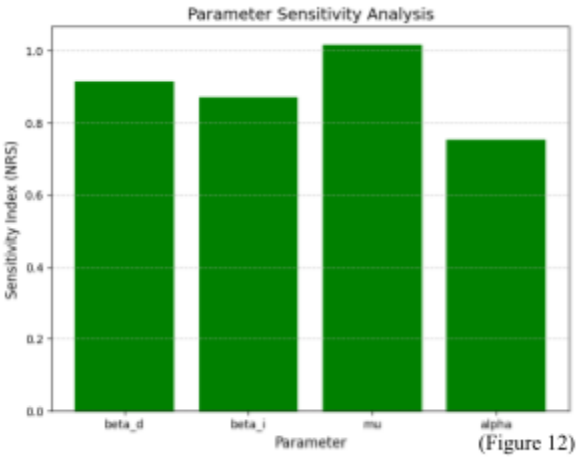


(Figure 9)

The contour plots show a deep decline in R_0 when μ is increased (Figure 8). Likewise, the threshold in the β_i vs. β_d comparison (Figure 9) highlights the key factors that interventions should be targeting in order to limit WNS persistence. The analysis of γ vs. μ (Figure 10) confirms that the pathogen decay rate must be considered, but host mortality has a stronger effect on R_0, emphasizing a tradeoff between reducing transmission and population stability. This clearly

shows us the importance and challenging aspect of balancing intervention strategies to minimize ecological harm while controlling the spread of the disease.

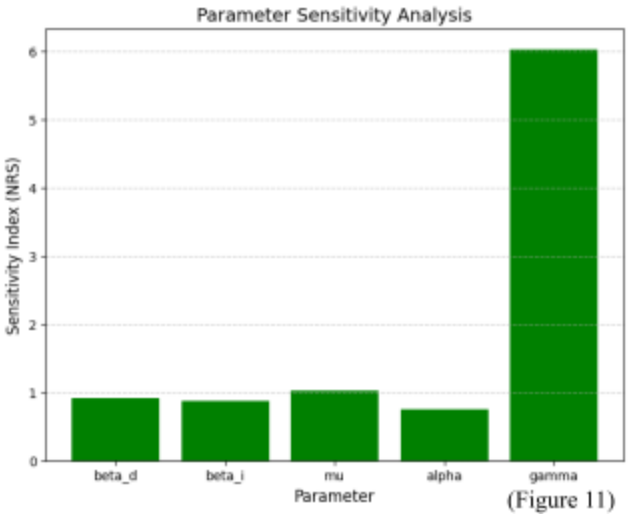
Normalized relative sensitivity (NRS) analysis



(Figure 12)

I performed a normalized relative sensitivity analysis to assess the changes in each parameter and their quantitative effect on R_0. Through this analysis, I measured the relative change in R_0 per unit change in each parameter, which allowed me to create a ranked assessment of the parameters that have the strongest influence on the WNS spread.

After conducting the analysis in the Python programming language, I found that the host mortality rate had the highest sensitivity index of 6.033, showing that increasing it is the most effective way to reduce R_0 (Figure 11). After omitting the host mortality rate (γ) and reproducing the bar graph (Figure 12), the pathogen decay was the second most sensitive parameter with a sensitivity index of 1.018, showing that an increase in the decay rate helps reduce R_0. Direct and indirect transmission rates had sensitivities at 0.914 and 0.872 respectively. The shedding rate had the lowest sensitivity index at 0.754, showing that a decrease in shedding helps reduce R_0 but it's certainly less impactful to the whole system than other parameters.



(Figure 11)

These results confirm and align with earlier one-at-a-time and multivariate analyses. The 3D surface plots showed that μ had a great local effect, and the NRS analysis shows us that, compared to the whole system, μ drives R_0 much more than other parameters (Figure 12), excluding host mortality rate. The ranking of these sensitivities shows that reducing pathogen decay (μ), the transmission rates (β_i and β_d), and maintaining a fair host mortality rate (γ), with caution because of ecological dismay, is really the most effective to control for effective WNS mitigation.

Discussion and Conclusion

Discussion

Through this study of the SIRP model, I'm able to provide a new understanding of how different parameters affect the spread of White-Nose Syndrome (WNS) in single-cave bat populations and the efficacy of various mitigation strategies. My research question was specifically designed to determine how varying transmission, shedding, mortality, and pathogen decay rates influence R_0 when R_0 is plotted as a function of said parameters. This, in turn, will allow for the assessment of which intervention strategies are most effective. Looking at the stress analyses, it's clear that an increase in host mortality and an increase in pathogen decay will have the greatest effect in lowering R_0. Despite the mathematical effect in reducing R_0, increasing host mortality poses ecological concerns, because of potential further declines in bat populations.

My results partially confirm my initial hypothesis that reducing transmission rates and increasing pathogen decay would be most effective. Despite the fact that reducing transmission rates does contribute to R_0, the impact of reducing them is less significant than initially proposed, as shown through the sensitivity analyses. This in part verifies prior assumptions that

environmental persistence is likely the primary driver; while my results do recognize that indirect transmission through the environment is a vital driver, the data show that it's certainly not the primary driver when compared to host mortality rate.

Another key understanding derived from my study is that increasing the rate of the pathogen decay helps, but it should not be done in isolation, because reducing transmission and environmental exposure have a similar effect on controlling the spread of the disease. In addition, studies conducted previously focused particularly on metapopulation dynamics or a specific cave characteristic, but I chose a much more controlled framework, where I'm able to explain the nuances of bats and the WNS non-linear dynamics rather than focusing solely on environmental decontamination.

Moreover, my results suggest that intervention strategies should aim to keep indirect transmission fairly low (transmission rate parameter below 0.25) and direct transmission even lower (below 0.21) to force R_0 below 1, thereby removing WNS from the cave completely. This threshold is a new, concrete guideline for the management of the WNS disease and its *Pseudogymnoascus destructans* pathogen. Also, I tried expanding on prior research and existing literature by applying various sensitivity methods, particularly the Normalized Relative Sensitivity approach that quantifies the importance of each parameter relative to one another (Ma et. al, 2022). The advantage of using this approach is to rank the importance of the parameters and thus the intervention strategies that correspond to those high-ranking parameters.

Conclusion

My study shows that in order to effectively mitigate WNS spread in bat populations, particularly in a single cave, a decrease in both direct and indirect transmission rates and a

balanced increase in host mortality and pathogen decay are necessary. Instead of increasing host mortality, which is the most effective strategy mathematically, my results suggest that remediation specialists should focus on targeting the pathogen and reducing bat-to-bat and environmental transmission rates when designing WNS control strategies. A possible combination could include bat vaccination to reduce their transmission and pesticide control to reduce the pathogen in the cave, which not only reduces the indirect transmission rate but also can increase the pathogen decay rate.

Some of the notable findings are the linear increase of R_0 with higher β_d, β_i, and α, and the exponential decay of R_0 with increasing γ and μ. Also, the substantiality of β_d and β_i was confirmed by the 3D surface and contour plots, but also their secondary role is clear, especially since μ remains more influential. Additionally, I was able to determine that γ is the most influential parameter in this particular SIRP system, followed by μ and the two β's.

The significance of my study is in the fact that it contributes to WNS management by offering a new, refined perspective to understand the dynamics of these parameters and the priorities that should be set. Unlike previous studies, which mostly emphasize environmental persistence, my research concludes that the problem is a bit more nuanced and may need a combination of mitigation and intervention strategies. Some of my results, however, do provide concrete numbers as thresholds for consideration in disease mitigation strategies among discussions between policymakers and conservationists.

However, as with any study, there are clear limitations. My model assumes that the bat populations are homogeneous, meaning it doesn't account for individual variation in susceptibility or biological factors that evoke varying responses. Additionally, my analysis doesn't incorporate spatial variation, through seasonal differences or particular weather

conditions, which certainly do affect the presence of the *P. destructans* pathogen, which thrives in fairly humid conditions. To further continue this discussion, future research should incorporate more complex ecological factors, like population heterogeneity, and consistent migration patterns within a certain bat population. These mathematical models can also be applied in other scenarios, including diseases within our human populations. Whether it be COVID-19 or monkeypox or other infectious diseases, we are able to construct similar models and perform similar analyses to guide public disease control. Likewise, they can be applied to other endangered species, helping us preserve the natural ecology of the earth.

The implications of these results are vast; it's relevant to wildlife disease managers, policymakers, and conservation biologists. Through my findings, I concluded that increasing pathogen decay and reducing transmission among bats through pesticide control and vaccination respectively will be most effective in eliminating WNS from a single, homogenous cave and bat population. As a result, future research should explore the feasibility and empirical efficacy of these interventions in real-world settings, further helping protect vulnerable bat populations from the deadly White-Nose Syndrome.

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Appendix A

```
# A public version of the code is available at
https://github.com/nazabu/WNS-Mathematical-Simulation


# The imports indicate that a new Python Jupyter Notebook was created
import numpy as np
import matplotlib.pyplot as plt


# Fixed parameters
alpha = 0.04   # Shedding rate of pathogen by infected hosts
mu = 0.05     # Pathogen decay rate
beta_d = 0.05  # Direct transmission rate
beta_i = 0.03  # Indirect transmission rate via environment


# Varying parameter (lower bound, higher bound, and resolution)
gamma_range = np.linspace(0.01, 0.5, 100)


R0_values = []


for gamma in gamma_range:
    R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
    R0_values.append(R0) # List of exact R0 values
```

```
# Plot
plt.figure(figsize=(10, 6))
plt.plot(gamma_range, R0_values, label=r'$R_0$', color='purple')
plt.axhline(y=1, color='r', linestyle='--', label=r'$R_0 = 1$ (Threshold)')
plt.xlabel(r'$\gamma$ (Host Mortality Rate)', fontsize=12)
plt.ylabel(r'$R_0$ (Basic Reproduction Number)', fontsize=12)
plt.title(r'$R_0$ as a Function of $\gamma$', fontsize=14)
plt.legend()
plt.grid()
plt.show()

print([tuple((round(float(gamma), 3), round(float(R0), 3))) for gamma, R0 in zip(gamma_range,
R0_values)])
```

```
import numpy as np
import matplotlib.pyplot as plt

# Fixed parameters
gamma = 0.5  # Host mortality rate
alpha = 0.04  # Shedding rate of pathogen by infected hosts
mu = 0.05  # Pathogen decay rate
```

```
beta_i = 0.03  # Direct transmission rate via environment


# Varying parameter
beta_d_range = np.linspace(0.01, 0.5, 100)  # lower bound, higher bound, and resolution


R0_values = []


for beta_d in beta_d_range:
    R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
    R0_values.append(R0)  # List of exact R0 values


# Plot
plt.figure(figsize=(10, 6))
plt.plot(beta_d_range, R0_values, label=r'$R_0$', color='b')
plt.axhline(y=1, color='r', linestyle='--', label=r'$R_0 = 1$ (Threshold)')
plt.xlabel(r'$β_d$ (Direct Transmission Rate)', fontsize=12)
plt.ylabel(r'$R_0$ (Basic Reproduction Number)', fontsize=12)
plt.title(r'$R_0$ as a Function of $β_d$', fontsize=14)
plt.legend()
plt.grid()
plt.show()
```

```
print([tuple((round(float(beta_d), 3), round(float(R0), 3))) for beta_d, R0 in zip(beta_d_range,
R0_values)])
```

```
import numpy as np
import matplotlib.pyplot as plt

# Fixed parameters
gamma = 0.5 # Host mortality rate
alpha = 0.04   # Shedding rate of pathogen by infected hosts
mu = 0.05     # Pathogen decay rate
beta_d = 0.03  # Indirect transmission rate via environment


# Varying parameter (lower bound, higher bound, and resolution)
beta_i_range = np.linspace(0.01, 0.5, 100)

R0_values = []

for beta_i in beta_i_range:
```

```
    R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
    R0_values.append(R0) # List of exact R0 values

# Plot
plt.figure(figsize=(10, 6))
plt.plot(beta_i_range, R0_values, label=r'$R_0$', color='y')
plt.axhline(y=1, color='r', linestyle='--', label=r'$R_0 = 1$ (Threshold)')
plt.xlabel(r'$β_i$ (Indirect Transmission Rate)', fontsize=12)
plt.ylabel(r'$R_0$ (Basic Reproduction Number)', fontsize=12)
plt.title(r'$R_0$ as a Function of $β_i$', fontsize=14)
plt.legend()
plt.grid()
plt.show()

print([tuple((round(float(beta_i), 3), round(float(R0), 3))) for beta_i, R0 in zip(beta_i_range,
R0_values)])
```

```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Fixed parameters
alpha = 0.04   # Shedding rate of pathogen by infected hosts
mu = 0.05     # Pathogen decay rate
beta_d = 0.05  # Direct transmission rate
beta_i = 0.03  # Indirect transmission rate via environment

# Varying parameter (lower bound, higher bound, and resolution)
gamma_range = np.linspace(0.01, 0.5, 100)

R0_values = []

for gamma in gamma_range:
    R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
    R0_values.append(R0) # List of exact R0 values


# Plot
plt.figure(figsize=(10, 6))
plt.plot(gamma_range, R0_values, label=r'$R_0$', color='purple')
plt.axhline(y=1, color='r', linestyle='--', label=r'$R_0 = 1$ (Threshold)')
plt.xlabel(r'$\gamma$ (Host Mortality Rate)', fontsize=12)
plt.ylabel(r'$R_0$ (Basic Reproduction Number)', fontsize=12)
plt.title(r'$R_0$ as a Function of $\gamma$', fontsize=14)
```

```
plt.legend()

plt.grid()

plt.show()


print([tuple((round(float(gamma), 3), round(float(R0), 3))) for gamma, R0 in zip(gamma_range,

R0_values)])




import numpy as np

import matplotlib.pyplot as plt


# Fixed parameters

gamma = 0.5 # Host mortality rate

alpha = 0.04   # Shedding rate of pathogen by infected hosts

beta_d = 0.05  # Direct transmission rate

beta_i = 0.03  # Indirect transmission rate via environment



# Varying parameter (lower bound, higher bound, and resolution)

mu_range = np.linspace(0.01, 0.5, 100)


R0_values = []
```

```
for mu in mu_range:
    R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
    R0_values.append(R0) # List of exact R0 values
# Plot
plt.figure(figsize=(10, 6))
plt.plot(mu_range, R0_values, label=r'$R_0$', color='g')
plt.axhline(y=1, color='r', linestyle='--', label=r'$R_0 = 1$ (Threshold)')
plt.xlabel(r'$\mu$ (Pathogen Decay Rate)', fontsize=12)
plt.ylabel(r'$R_0$ (Basic Reproduction Number)', fontsize=12)
plt.title(r'$R_0$ as a Function of $\mu$', fontsize=14)
plt.legend()
plt.grid()
plt.show()

print([tuple((round(float(mu), 3), round(float(R0), 3))) for mu, R0 in zip(mu_range, R0_values)])
```

```
from R0_values import *
import matplotlib.pyplot as plt

# Define parameter ranges
```

```
param_ranges = {
    "beta_d": np.linspace(0.01, 0.5, 100),
    "beta_i": np.linspace(0.01, 0.5, 100),
    "mu": np.linspace(0.01, 0.5, 100),
    "alpha": np.linspace(0.01, 0.5, 100),
    "gamma": np.linspace(0.01, 0.5, 100)
}

# R0 ranges for each parameter
R0_ranges = {
    "beta_d": beta_d_R0_values(),
    "beta_i": beta_i_R0_values(),
    "mu": mu_R0_values(),
    "alpha": alpha_R0_values(),
    "gamma": gamma_R0_values(),
}

# Compute sensitivity index for each parameter (NRS)
sensitivity = {}
for param in param_ranges.keys():
    delta_X = np.max(param_ranges[param]) - np.min(param_ranges[param])
    delta_R0 = np.max(R0_ranges[param]) - np.min(R0_ranges[param])
```

```
    X_avg = np.mean(param_ranges[param])
    R0_avg = np.mean(R0_ranges[param])

    S_X = (delta_R0 / R0_avg) / (delta_X / X_avg)
    sensitivity[param] = S_X

#del sensitivity["gamma"] # can also delete gamma (bc it's so large) to scale the graph more
properly

# Sorting params
sorted_sensitivity = sorted(sensitivity.items(), key=lambda x: x[1], reverse=True)

for param, S_X in sorted_sensitivity:
    print(f"Parameter: {param}, Sensitivity Index: {S_X:.3f}")

plt.figure(figsize=(8, 6))
plt.bar(sensitivity.keys(), sensitivity.values(), color='green')
plt.xlabel("Parameter", fontsize=12)
plt.ylabel("Sensitivity Index (NRS)", fontsize=12)
plt.title("Parameter Sensitivity Analysis", fontsize=14)
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.show()
```

```
import numpy as np

'''Alpha R0 Values'''
def alpha_R0_values():
    # Fixed parameters
    gamma = 0.5 # Host mortality rate
    mu = 0.05     # Pathogen decay rate
    beta_d = 0.05  # Direct transmission rate
    beta_i = 0.03  # Indirect transmission rate via environment

    # Varying parameter (lower bound, higher bound, and resolution)
    alpha_range = np.linspace(0.01, 0.5, 100)

    R0_values = []
    for alpha in alpha_range:
        R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
        R0_values.append(R0) # List of exact R0 values for alpha

    return R0_values
```

```
'''Beta_D R0 Values'''
def beta_d_R0_values():
    # Fixed parameters
    gamma = 0.5  # Host mortality rate
    alpha = 0.04  # Shedding rate of pathogen by infected hosts
    mu = 0.05  # Pathogen decay rate
    beta_i = 0.03  # Direct transmission rate via environment

    # Varying parameter
    beta_d_range = np.linspace(0.01, 0.5, 100)  # lower bound, higher bound, and resolution

    R0_values = []

    for beta_d in beta_d_range:
        R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
        R0_values.append(R0)  # List of exact R0 values

    return R0_values
```

```
'''Beta_I R0 Values'''
def beta_i_R0_values():
    # Fixed parameters
    gamma = 0.5 # Host mortality rate
    alpha = 0.04   # Shedding rate of pathogen by infected hosts
    mu = 0.05     # Pathogen decay rate
    beta_d = 0.03  # Indirect transmission rate via environment



    # Varying parameter (lower bound, higher bound, and resolution)
    beta_i_range = np.linspace(0.01, 0.5, 100)

    R0_values = []

    for beta_i in beta_i_range:
        R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
        R0_values.append(R0) # List of exact R0 values

    return R0_values
```

```
'''Gamma R0 Values'''
def gamma_R0_values():
    # Fixed parameters
    alpha = 0.04   # Shedding rate of pathogen by infected hosts
    mu = 0.05     # Pathogen decay rate
    beta_d = 0.05  # Direct transmission rate
    beta_i = 0.03  # Indirect transmission rate via environment

    # Varying parameter (lower bound, higher bound, and resolution)
    gamma_range = np.linspace(0.01, 0.5, 100)

    R0_values = []

    for gamma in gamma_range:
        R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
        R0_values.append(R0) # List of exact R0 values

    return R0_values
```

```
'''Mu R0 Values'''
def mu_R0_values():
    # Fixed parameters
    gamma = 0.5 # Host mortality rate
    alpha = 0.04   # Shedding rate of pathogen by infected hosts
    beta_d = 0.05  # Direct transmission rate
    beta_i = 0.03  # Indirect transmission rate via environment

    # Varying parameter (lower bound, higher bound, and resolution)
    mu_range = np.linspace(0.01, 0.5, 100)

    R0_values = []

    for mu in mu_range:
        R0 = (beta_d / gamma) + (beta_i / gamma) * (alpha / mu)
        R0_values.append(float(R0)) # List of exact R0 values

    return R0_values

import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt
```

```
# Quadratic solver
def quadratic(a, b, c):
    x1 = (-b + np.sqrt(b**2 - 4*a*c)) / (2*a)
    #x2 = (-b - np.sqrt(b**2 - 4*a*c)) / (2*a)
    return x1 #, x2

# SIRP model differential equations
def SIRP_model(y, t, N, beta_d, beta_i, gamma, alpha, mu):
    S, I, R, P = y

    dSdt = -beta_d * S * I / N - beta_i * S * P
    dIdt = beta_d * S * I / N + beta_i * S * P - gamma * I
    dRdt = gamma * I
    dPdt = alpha * I - mu * P

    return [dSdt, dIdt, dRdt, dPdt]

# Parameters
beta_d = 0.02  # Direct transmission rate
beta_i = 0.01  # Indirect transmission rate via environment
gamma = 0.01  # Host mortality rate
alpha = 0.01  # Shedding rate of pathogen by infected hosts
mu = 0.05  # Pathogen decay rate
```

```
# Initial conditions
S0 = 990  # Initial number of susceptible individuals
I0 = 10  # Initial number of infected individuals
R0 = 0  # Initial number of removed individuals
P0 = 1  # Initial pathogen load in the environment
N = S0 + I0 # Total population

y0 = [S0, I0, R0, P0]

# Time points
t = np.linspace(0, 100, 100)  # Simulate for 100 days with 100 time steps

# Solve the differential equations
ret = odeint(SIRP_model, y0, t, args=(N, beta_d, beta_i, gamma, alpha, mu))

# Get solutions
S, I, R, P = ret.T

# Solve for R0
# a = 1
# b = (-1/gamma) * (beta_d + (beta_i * alpha) / mu)
# c = (-beta_i * 1) / mu
```

```
R0 = (beta_d / gamma) + ((beta_i / gamma) * (alpha / mu))

# R0 = (1 / 2) * (beta_d / gamma + np.sqrt((beta_d / gamma)**2 + (4 * beta_i * alpha) / (mu *
gamma))) faulty R0

plt.figure(figsize=(12, 8))
plt.plot(t, S, label='Susceptible (S)')
plt.plot(t, I, label='Infected (I)')
plt.plot(t, R, label='Removed (R)')
plt.plot(t, P, label='Pathogen Load (P)', linestyle='--')
plt.xlabel('Time (days)')
plt.ylabel('Population / Pathogen Load')
plt.title(f"SIRP Model (R0 = {R0:.2f})")
plt.legend()
plt.grid()
plt.show()
```

```
import matplotlib.pyplot as plt
import numpy as np


def compute_R0(beta_d, gamma, beta_i, alpha, mu):
    R0 = (beta_d / gamma) + ((beta_i / gamma) * (alpha / mu))
    return R0


beta_d_range = np.linspace(0.01, 0.05, 100)
mu_range = np.linspace(0.01, 0.05, 100)



beta_d_mesh, mu_mesh = np.meshgrid(beta_d_range, mu_range)


R0_mesh = compute_R0(beta_d_mesh, 0.03, 0.5, 0.04, mu_mesh)



plt.figure(figsize=(8,6))
contour = plt.contourf(beta_d_mesh, mu_mesh, R0_mesh, levels=50, cmap="coolwarm")
plt.colorbar(contour, label="R0 Value")
plt.xlabel("Direct Transmission Rate (β_d)")
plt.ylabel("Pathogen Decay Rate (μ)")
plt.title("Contour Plot of R0 as a Function of β_d and μ")
plt.show()
```

```
# Ensure level values are strictly increasing
if R0_mesh.min() < 1 < R0_mesh.max():
    plt.contourf(beta_d_mesh, mu_mesh, R0_mesh, levels=np.linspace(R0_mesh.min(), 1 - 1e-6,
50), colors='blue', alpha=0.5)  # Below R0=1
    plt.contourf(beta_d_mesh, mu_mesh, R0_mesh, levels=np.linspace(1 + 1e-6, R0_mesh.max(),
50), colors='red', alpha=0.5)  # Above R0=1



fig = plt.figure(figsize=(10,7))
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(beta_d_mesh, mu_mesh, R0_mesh, cmap="viridis")

Z_plane = np.ones_like(R0_mesh)
ax.plot_surface(beta_d_mesh, mu_mesh, Z_plane, color='red', alpha=0.5)

ax.set_xlabel("β_d")
ax.set_ylabel("μ")
ax.set_zlabel("R0")
plt.title("3D Surface Plot of R0")
ax.view_init(elev=30., azim=35)
plt.show()
```

```
beta_d_range = np.linspace(0.01, 0.05, 100)
beta_i_range = np.linspace(0.01, 0.05, 100)

beta_d_mesh, beta_i_mesh = np.meshgrid(beta_d_range, beta_i_range)

R0_mesh = compute_R0(beta_d_mesh, 0.03, beta_i_mesh, 0.04, 0.05)

plt.figure(figsize=(8,6))
contour = plt.contourf(beta_d_mesh, beta_i_mesh, R0_mesh, levels=50, cmap="coolwarm")
plt.colorbar(contour, label="R0 Value")
plt.xlabel("Direct Transmission Rate (β_d)")
plt.ylabel("Indirect transmission rate (β_i)")
plt.title("Contour Plot of R0 as a Function of β_d and β_i")
plt.show()


# Ensure level values are strictly increasing
if R0_mesh.min() < 1 < R0_mesh.max():
    plt.contourf(beta_d_mesh, beta_i_mesh, R0_mesh, levels=np.linspace(R0_mesh.min(), 1 -
1e-6, 50), colors='blue', alpha=0.5)  # Below R0=1
```

```
    plt.contourf(beta_d_mesh, beta_i_mesh, R0_mesh, levels=np.linspace(1 + 1e-6,
R0_mesh.max(), 50), colors='red', alpha=0.5)  # Above R0=1



fig = plt.figure(figsize=(10,7))
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(beta_d_mesh, beta_i_mesh, R0_mesh, cmap="viridis")

Z_plane = np.ones_like(R0_mesh)
ax.plot_surface(beta_d_mesh, beta_i_mesh, Z_plane, color='red', alpha=0.5)

ax.set_xlabel("β_d")
ax.set_ylabel("β_i")
ax.set_zlabel("R0")
plt.title("3D Surface Plot of R0")
plt.show()



mu_range = np.linspace(0.01, 0.5, 100)
gamma_range = np.linspace(0.01, 0.5, 100)

mu_mesh, gamma_mesh = np.meshgrid(mu_range, gamma_range)
```

```
R0_mesh = compute_R0(0.05, gamma_mesh, 0.03, 0.04, mu_mesh)


plt.figure(figsize=(8,6))
contour = plt.contourf(mu_mesh, gamma_mesh, R0_mesh, levels=50, cmap="coolwarm")
plt.colorbar(contour, label="R0 Value")
plt.xlabel("Pathogen Decay Rate (μ)")
plt.ylabel("Host Mortality Rate (γ)")
plt.title("Contour Plot of R0 as a Function of μ and γ")
plt.show()



# Ensure level values are strictly increasing
if R0_mesh.min() < 1 < R0_mesh.max():
    plt.contourf(mu_mesh, gamma_mesh, R0_mesh, levels=np.linspace(R0_mesh.min(), 1 - 1e-6,
50), colors='blue', alpha=0.5)  # Below R0=1
    plt.contourf(mu_mesh, gamma_mesh, R0_mesh, levels=np.linspace(1 + 1e-6,
R0_mesh.max(), 50), colors='red', alpha=0.5)  # Above R0=1



fig = plt.figure(figsize=(10,7))
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(mu_mesh, gamma_mesh, R0_mesh, cmap="viridis")
```

```
Z_plane = np.ones_like(R0_mesh)
ax.plot_surface(mu_mesh, gamma_mesh, Z_plane, color='red', alpha=0.75)

ax.set_xlabel("μ")
ax.set_ylabel("γ")
ax.set_zlabel("R0")
plt.title("3D Surface Plot of R0")
ax.view_init(elev=30., azim=40)
plt.show()
```